



Functional Requirement Document

EMERGENCY DESCENT function

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1 Executive summary

This document is the functional specification of the EMERGENCY DESCENT function. It provides “aircraft-level” requirements that will be the main input for systems specifications.

The purpose of the EMERGENCY DESCENT function is to have the aircraft descend to a safe altitude in the case of loss of cabin pressurisation.

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2 Record of revisions

Issue	DATE	Reason for revision
V01	09/03/2026	First issue

3 General

3.1 Objective

This document is the functional specification for the EMERGENCY DESCENT function.

- Section 4 provides a high-level description of the EMERGENCY DESCENT function.
- Section 5 describes operational scenarios of the EMERGENCY DESCENT function.
- Section 6 contains "aircraft level" requirements for the EMERGENCY DESCENT function.

3.2 Applicability

[1] is the baseline for this functional specification. Thus this document applies to any aircraft described by [1].

3.3 Reference documents

[1] Baseline avionics description.

[2] Arinc 702a-5: Advanced flight management computer system. 2018.

3.4 Acronyms

AP	Autopilot
ND	Navigation Display
FCU	Flight Control Unit
IAS	Indicated Air Speed
MSA	Minimum Safe Altitude

3.5 Definitions

4 Function Description

The purpose of the EMERGENCY DESCENT function is to navigate the aircraft from its current altitude to 10 000 ft or the MSA in the current sector, following the fastest descent profile.

The EMERGENCY DESCENT function provides the following features:

- It allows the flight crew to activate the EMERGENCY DESCENT.
- Upon activation of the EMERGENCY DESCENT:

The aircraft begins descending to 10 000 ft at 250 kts IAS.

It allows the flight crew to select a desired altitude if different from 10 000 ft.

It allows the flight crew to set a desired IAS for the descent.

- Upon arriving to the selected altitude, the aircraft levels off.
- It allows the flight crew to deactivate the EMERGENCY DESCENT, in which case the aircraft levels off at its current altitude.

Note: The flight crew is responsible for extending the airbrakes to increase the vertical speed of the descent. They are also responsible for the retraction of the airbrakes when the aircraft levels off.

5 Operational Scenarios

6 Requirements

7 Open items

8 Appendices